

ARAR HOSPITAL Fire Sprinkler Reports

for

Arar (Sprinkler Calculation)

Demand

Prepared By:

OMAR ISMAIL

OMAR ISMAIL

OCTOBER 05,2009



General Project Data Report

General Data

Project Title:	Arar HOSPITAL	Project File Name:	ARAR HOSPITAL.fiw
Designed By:	OMAR ISMAIL	Date:	OCTOBER 05,2009
Code Reference:	NFPA 13	Approving Agency:	
Client Name:	Arar (Sprinkler Calculation)	Phone:	
Address:	Demand	City, State Zip Code:	
Company Name:	OMAR ISMAIL	Representative:	OMAR ISMAIL
Company Address:		City And State:	
Phone:			
Building Name:	Arar (Sprinkler Calculation)	Building Owner:	
Contact at Building:		Phone at Building:	
Address Of Building:		City, State Zip Code:	

Project Data

Description Of Hazard:	Ordinary 1	Sprinkler System Type:	Wet
Design Area Of Water Application:	2016 ft ²	Maximum Area Per Sprinkler:	150 ft ²
Default Sprinkler K-Factor:	5.80 K	Default Pipe Material:	SCHED 40 WET STEEL
Inside Hose Stream Allowance:	0.00 gpm	Outside Hose Stream Allowance:	0.00 gpm
In Rack Sprinkler Allowance:	0.00 gpm		

Sprinkler Specifications

Make:	GLOBE	Model:	G
Size:	1	Temperature Rating:	160 F

Water Supply Test Data

Source Of Information:		Date Of Test:	
Test Hydrant ID:			
Hydrant Elevation:	0 ft	Static Pressure:	0.00 psi
Test Flow Rate:	0.00 gpm	Test Residual Pressure:	0.00 psi
Calculated System Flow Rate:	740.73 gpm	Calculated Inflow Residual Pressure:	126.31 psi

Calculation Project Data

Calculation Mode:	Demand		
HMD Minimum Residual Pressure:	7.00 psi	Minimum Desired Flow Density:	0.15 gpm/ft ²
Number Of Active Nodes:	67		
Number Of Active Pipes:	66	Number Of Inactive Pipes:	0
Number Of Active Sprinklers:	24	Number Of Inactive Sprinklers:	0



Fire Sprinkler Input Data

Node Input Data

Node No.	Node Description	Area Group	Sprinkler KFactor (K)	Pressure Estimate (psi)	Node Elevation (feet)	Non-Sprinkler Discharge (gpm)
1	No Discharge		N/A	126.31	0.00	0.00
2	No Discharge		N/A	125.85	0.00	0.00
3	No Discharge		N/A	127.04	-3.00	0.00
4	No Discharge		N/A	126.30	-3.00	0.00
5	No Discharge		N/A	103.41	49.00	0.00
7	No Discharge		N/A	103.09	49.00	0.00
8	No Discharge		N/A	98.93	49.00	0.00
9	No Discharge		N/A	84.50	49.00	0.00
10	No Discharge		N/A	64.44	49.00	0.00
11	No Discharge		N/A	51.19	49.00	0.00
12	No Discharge		N/A	45.38	49.00	0.00
13	No Discharge		N/A	36.60	49.00	0.00
14	No Discharge		N/A	30.28	49.00	0.00
15	No Discharge		N/A	27.69	49.00	0.00
16	No Discharge		N/A	23.94	49.00	0.00
60	No Discharge		N/A	103.21	49.00	0.00
61	No Discharge		N/A	100.33	49.00	0.00
62	No Discharge		N/A	94.79	49.00	0.00
63	No Discharge		N/A	72.27	49.00	0.00
64	Non-Sprinkler		N/A	69.16	43.00	100.00
100	No Discharge		N/A	54.28	50.00	0.00
101	No Discharge		N/A	32.71	50.00	0.00
102	Sprinkler		5.80	21.18	49.00	0.00
103	No Discharge		N/A	30.82	50.00	0.00
104	Sprinkler		5.80	30.94	49.00	0.00
105	No Discharge		N/A	50.67	50.00	0.00



Fire Sprinkler Input Data

Node Input Data (cont'd)

Node No.	Node Description	Area Group	Sprinkler KFactor (K)	Pressure Estimate (psi)	Node Elevation (feet)	Non-Sprinkler Discharge (gpm)
106	Sprinkler		5.80	50.61	49.00	0.00
107	No Discharge		N/A	34.03	50.00	0.00
108	Sprinkler		5.80	34.12	49.00	0.00
109	Sprinkler		5.80	30.99	49.00	0.00
110	No Discharge		N/A	22.74	50.00	0.00
111	No Discharge		N/A	49.06	50.00	0.00
112	No Discharge		N/A	45.67	50.00	0.00
113	Sprinkler		5.80	45.65	49.00	0.00
114	No Discharge		N/A	32.01	50.00	0.00
115	Sprinkler		5.80	32.12	49.00	0.00
116	Sprinkler		5.80	30.00	49.00	0.00
117	Sprinkler		5.80	22.94	49.00	0.00
120	No Discharge		N/A	40.50	50.00	0.00
121	No Discharge		N/A	24.18	50.00	0.00
122	Sprinkler		5.80	23.18	49.00	0.00
123	Sprinkler		5.80	15.34	49.00	0.00
124	No Discharge		N/A	16.72	50.00	0.00
125	Sprinkler		5.80	16.97	49.00	0.00
130	No Discharge		N/A	33.32	50.00	0.00
131	No Discharge		N/A	26.69	50.00	0.00
132	Sprinkler		5.80	26.85	49.00	0.00
133	Sprinkler		5.80	24.14	49.00	0.00
140	No Discharge		N/A	26.84	50.00	0.00
141	No Discharge		N/A	15.83	50.00	0.00
142	Sprinkler		5.80	15.29	49.00	0.00
143	Sprinkler		5.80	21.18	49.00	0.00
144	No Discharge		N/A	10.81	50.00	0.00



Fire Sprinkler Input Data

Node Input Data (cont'd)

Node No.	Node Description	Area Group	Sprinkler KFactor (K)	Pressure Estimate (psi)	Node Elevation (feet)	Non-Sprinkler Discharge (gpm)
145	Sprinkler		5.80	11.12	49.00	0.00
150	No Discharge		N/A	25.08	50.00	0.00
151	No Discharge		N/A	20.03	50.00	0.00
152	Sprinkler		5.80	20.25	49.00	0.00
153	Sprinkler		5.80	17.97	49.00	0.00
160	No Discharge		N/A	20.67	50.00	0.00
161	No Discharge		N/A	12.58	50.00	0.00
162	Sprinkler		5.80	12.21	49.00	0.00
163	Sprinkler		5.80	21.18	49.00	0.00
164	No Discharge		N/A	15.93	50.00	0.00
165	Sprinkler		5.80	16.19	49.00	0.00
166	Sprinkler		5.80	14.64	49.00	0.00
167	No Discharge		N/A	8.52	50.00	0.00
168	Sprinkler		5.80	21.18	49.00	0.00



Fire Sprinkler Input Data

Pipe Input Data

Beg. Node	End. Node	Pipe Description	Nominal Diameter (inch)	Type Group	Fitting Data	Nominal Length (feet)	Fitting Length (feet)	Total Length (feet)	CFactor (gpm/inch-psi)
1	2	SCHED 40 WET STEEL	8.000	8	EBGC	6.60	79.00	85.60	120
2	3	SCHED 40 WET STEEL	8.000	8	E	3.00	18.00	21.00	120
3	4	SCHED 40 WET STEEL	8.000	8	E	122.00	18.00	140.00	120
4	5	SCHED 40 WET STEEL	8.000	8	E	52.00	18.00	70.00	120
5	60	SCHED 40 WET STEEL	8.000	8	T	2.00	35.00	37.00	120
60	7	SCHED 40 WET STEEL	8.000	8	EG	8.00	22.00	30.00	120
7	8	SCHED 40 WET STEEL	5.000	5	2E	86.00	24.00	110.00	120
8	9	SCHED 40 WET STEEL	4.000	4		127.00	0.00	127.00	120
9	10	SCHED 40 WET STEEL	3.000	3	ET	25.00	22.00	47.00	120
10	100	SCHED 40 WET STEEL	1.500	9	T	1.00	8.00	9.00	120
10	11	SCHED 40 WET STEEL	2.500	3	T	9.00	12.00	21.00	120
11	111	SCHED 40 WET STEEL	1.500	9	E	1.00	4.00	5.00	120
11	12	SCHED 40 WET STEEL	2.500	3	T	3.00	12.00	15.00	120
12	120	SCHED 40 WET STEEL	1.000	1	E	1.00	2.00	3.00	120
12	13	SCHED 40 WET STEEL	2.000	2	T	5.00	10.00	15.00	120
13	130	SCHED 40 WET STEEL	1.000	1	E	1.00	2.00	3.00	120
13	14	SCHED 40 WET STEEL	2.000	2	T	7.00	10.00	17.00	120
14	140	SCHED 40 WET STEEL	1.000	1	E	1.00	2.00	3.00	120
14	15	SCHED 40 WET STEEL	2.000	2	T	3.00	10.00	13.00	120
15	150	SCHED 40 WET STEEL	1.000	1	E	1.00	2.00	3.00	120
15	16	SCHED 40 WET STEEL	1.500	9	E	8.00	4.00	12.00	120
16	160	SCHED 40 WET STEEL	1.500	9	T	1.00	8.00	9.00	120
60	61	SCHED 40 WET STEEL	2.500	3		73.00	0.00	73.00	120
61	62	SCHED 40 WET STEEL	2.000	2	E	54.00	5.00	59.00	120
62	63	SCHED 40 WET STEEL	1.500	9	E	67.00	4.00	71.00	120
63	64	SCHED 40 WET STEEL	1.500	9	2E	10.00	8.00	18.00	120



Fire Sprinkler Input Data

Pipe Input Data (cont'd)

Beg. Node	End. Node	Pipe Description	Nominal Diameter (inch)	Type Group	Fitting Data	Nominal Length (feet)	Fitting Length (feet)	Total Length (feet)	CFactor (gpm/inch-psi)
100	101	SCHED 40 WET STEEL	1.000	1	T	6.00	5.00	11.00	120
100	105	SCHED 40 WET STEEL	1.500	9	T	2.00	8.00	10.00	120
101	103	SCHED 40 WET STEEL	1.000	1	E	4.00	2.00	6.00	120
101	110	SCHED 40 WET STEEL	1.000	1	T	7.00	5.00	12.00	120
110	102	SCHED 40 WET STEEL	1.000	1	E	7.00	2.00	9.00	120
103	104	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
105	106	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
105	107	SCHED 40 WET STEEL	1.000	1	T	9.00	5.00	14.00	120
107	108	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
107	109	SCHED 40 WET STEEL	1.000	1	E	9.00	2.00	11.00	120
110	117	SCHED 40 WET STEEL	1.000	0		1.00	0.00	1.00	120
111	112	SCHED 40 WET STEEL	1.500	9	T	2.00	8.00	10.00	120
112	113	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
112	114	SCHED 40 WET STEEL	1.000	1	T	7.00	5.00	12.00	120
114	115	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
114	116	SCHED 40 WET STEEL	1.000	1	E	6.00	2.00	8.00	120
120	121	SCHED 40 WET STEEL	1.000	1	T	6.00	5.00	11.00	120
121	122	SCHED 40 WET STEEL	1.000	1	E	4.00	2.00	6.00	120
121	124	SCHED 40 WET STEEL	1.000	1	T	7.00	5.00	12.00	120
124	123	SCHED 40 WET STEEL	1.000	1	E	9.00	2.00	11.00	120
124	125	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
130	131	SCHED 40 WET STEEL	1.000	1	T	2.00	5.00	7.00	120
131	132	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
131	133	SCHED 40 WET STEEL	1.000	1	E	10.00	2.00	12.00	120
140	141	SCHED 40 WET STEEL	1.000	1	T	6.00	5.00	11.00	120
141	142	SCHED 40 WET STEEL	1.000	1	E	4.00	2.00	6.00	120
141	144	SCHED 40 WET STEEL	1.000	1	T	7.00	5.00	12.00	120



Fire Sprinkler Input Data

Pipe Input Data (cont'd)

Beg. Node	End. Node	Pipe Description	Nominal Diameter (inch)	Type Group	Fitting Data	Nominal Length (feet)	Fitting Length (feet)	Total Length (feet)	CFactor (gpm/inch-psi)
144	143	SCHED 40 WET STEEL	1.000	1	E	10.00	2.00	12.00	120
144	145	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
150	151	SCHED 40 WET STEEL	1.000	1	T	2.00	5.00	7.00	120
151	152	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
151	153	SCHED 40 WET STEEL	1.000	1	2E	9.00	4.00	13.00	120
160	161	SCHED 40 WET STEEL	1.000	1	E	8.00	2.00	10.00	120
160	164	SCHED 40 WET STEEL	1.000	1	T	3.00	5.00	8.00	120
161	162	SCHED 40 WET STEEL	1.000	1	E	4.00	2.00	6.00	120
161	167	SCHED 40 WET STEEL	1.000	1	T	7.00	5.00	12.00	120
167	163	SCHED 40 WET STEEL	1.000	1	E	10.00	2.00	12.00	120
164	165	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120
164	166	SCHED 40 WET STEEL	1.000	1	E	9.00	2.00	11.00	120
167	168	SCHED 40 WET STEEL	1.000	1		1.00	0.00	1.00	120



Fire Sprinkler Output Data

Group Peak Node Groupings Output Data

Pipe Segment Beg. Node	Pipe Segment End. Node	Pipe Type Group	Pipe Flow Rate (gpm)	Sprinkler Flow At Beg. Node (gpm)	Non-Sprinkler Flow Out (+) (gpm)	Non-Sprinkler Flow In (-) (gpm)	Beg. Node Residual Pressure (psi)	Imbalance Flow At Beg. Node (gpm)
1	2	8	740.73	0.00	0.00	-740.73	126.31	
2	1	8	-740.73	0.00	0.00	0.00	125.85	0.02330
2	3	8	740.76					
3	2	8	-740.76	0.00	0.00	0.00	127.04	-0.03273
3	4	8	740.72					
4	3	8	-740.72	0.00	0.00	0.00	126.30	0.01130
4	5	8	740.74					
5	4	8	-740.74	0.00	0.00	0.00	103.41	0.01027
5	60	8	740.75					
7	60	8	-640.72	0.00	0.00	0.00	103.09	-0.05216
7	8	5	640.66					
8	7	5	-640.66	0.00	0.00	0.00	98.93	-0.01772
8	9	4	640.65					
9	8	4	-640.65	0.00	0.00	0.00	84.50	-0.00485
9	10	3	640.64					
10	9	3	-640.64	0.00	0.00	0.00	64.44	-0.03078
10	100	9	193.86					
10	11	3	446.75					
11	10	3	-446.75	0.00	0.00	0.00	51.19	-0.00077
11	111	9	103.60					
11	12	3	343.14					
12	11	3	-343.14	0.00	0.00	0.00	45.38	-0.00530
12	120	1	74.49					
12	13	2	268.65					
13	12	2	-268.65	0.00	0.00	0.00	36.60	0.06044
13	130	1	58.50					
13	14	2	210.21					
14	13	2	-210.21	0.00	0.00	0.00	30.28	0.12626
14	140	1	60.28					
14	15	2	150.05					
15	14	2	-150.05	0.00	0.00	0.00	27.69	-0.18403
15	150	1	50.67					
15	16	9	99.19					
16	15	9	-99.19	0.00	0.00	0.00	23.94	0.34286
16	160	9	99.53					
60	5	8	-740.75	0.00	0.00	0.00	103.21	-0.07588
60	7	8	640.72					



Fire Sprinkler Output Data

Group Peak Node Groupings Output Data

Pipe Segment Beg. Node	End. Node	Pipe Type Group	Pipe Flow Rate (gpm)	Sprinkler Flow At Beg. Node (gpm)	Non-Sprinkler Flow Out (+) (gpm)	In (-) (gpm)	Beg. Node Residual Pressure (psi)	Imbalance Flow At Beg. Node (gpm)
60	61	3	99.95					
61	60	3	-99.95	0.00	0.00	0.00	100.33	-0.00152
61	62	2	99.95					
62	61	2	-99.95	0.00	0.00	0.00	94.79	-0.00323
62	63	9	99.95					
63	62	9	-99.95	0.00	0.00	0.00	72.27	0.00316
63	64	9	99.95					
64	63	9	-99.95	0.00	100.00	0.00	69.16	0.04845
100	10	9	-193.86	0.00	0.00	0.00	54.28	-0.02060
100	101	1	86.65					
100	105	9	107.20					
101	100	1	-86.65	0.00	0.00	0.00	32.71	-0.00652
101	103	1	32.20					
101	110	1	54.44					
102	110	1	-26.69	26.69	0.00	0.00	21.18	-0.00176
103	101	1	-32.20	0.00	0.00	0.00	30.82	-0.00006
103	104	1	32.20					
104	103	1	-32.20	32.26	0.00	0.00	30.94	0.06095
105	100	9	-107.20	0.00	0.00	0.00	50.67	-0.00493
105	106	1	41.09					
105	107	1	66.10					
106	105	1	-41.09	41.26	0.00	0.00	50.61	0.17204
107	105	1	-66.10	0.00	0.00	0.00	34.03	-0.08211
107	108	1	33.77					
107	109	1	32.25					
108	107	1	-33.77	33.88	0.00	0.00	34.12	0.11317
109	107	1	-32.25	32.29	0.00	0.00	30.99	0.03713
110	101	1	-54.44	0.00	0.00	0.00	22.74	0.00005
110	102	1	26.69					
110	117	0	27.74					
111	11	9	-103.60	0.00	0.00	0.00	49.06	-0.00292
111	112	9	103.60					
112	111	9	-103.60	0.00	0.00	0.00	45.67	-0.01290
112	113	1	39.03					
112	114	1	64.56					



Fire Sprinkler Output Data

Group Peak Node Groupings Output Data

Pipe Segment Beg. Node	End. Node	Pipe Type Group	Pipe Flow Rate (gpm)	Sprinkler Flow At Beg. Node (gpm)	Non-Sprinkler Flow Out (+) (gpm)	In (-) (gpm)	Beg. Node Residual Pressure (psi)	Imbalance Flow At Beg. Node (gpm)
113	112	1	-39.03	39.19	0.00	0.00	45.65	0.15900
114	112	1	-64.56	0.00	0.00	0.00	32.01	0.11880
114	115	1	32.95					
114	116	1	31.73					
115	114	1	-32.95	32.87	0.00	0.00	32.12	-0.07810
116	114	1	-31.73	31.77	0.00	0.00	30.00	0.03876
117	110	0	-27.74	27.78	0.00	0.00	22.94	0.03422
120	12	1	-74.49	0.00	0.00	0.00	40.50	0.00070
120	121	1	74.49					
121	120	1	-74.49	0.00	0.00	0.00	24.18	-0.09320
121	122	1	27.82					
121	124	1	46.58					
122	121	1	-27.82	27.92	0.00	0.00	23.18	0.10524
123	124	1	-22.68	22.72	0.00	0.00	15.34	0.03995
124	121	1	-46.58	0.00	0.00	0.00	16.72	-0.07749
124	123	1	22.68					
124	125	1	23.82					
125	124	1	-23.82	23.89	0.00	0.00	16.97	0.07175
130	13	1	-58.50	0.00	0.00	0.00	33.32	-0.01829
130	131	1	58.49					
131	130	1	-58.49	0.00	0.00	0.00	26.69	-0.32275
131	132	1	29.82					
131	133	1	28.34					
132	131	1	-29.82	30.05	0.00	0.00	26.85	0.23130
133	131	1	-28.34	28.50	0.00	0.00	24.14	0.15476
140	14	1	-60.28	0.00	0.00	0.00	26.84	-0.04908
140	141	1	60.24					
141	140	1	-60.24	0.00	0.00	0.00	15.83	-0.11879
141	142	1	22.54					
141	144	1	37.58					
142	141	1	-22.54	22.68	0.00	0.00	15.29	0.14056
143	144	1	-18.25	26.69	0.00	0.00	21.18	8.44300
144	141	1	-37.58	0.00	0.00	0.00	10.81	-0.11970
144	143	1	18.25					



Fire Sprinkler Output Data

Group Peak Node Groupings Output Data

Pipe Segment Beg. Node	End. Node	Pipe Type Group	Pipe Flow Rate (gpm)	Sprinkler Flow At Beg. Node (gpm)	Non-Sprinkler Flow Out (+) (gpm)	In (-) (gpm)	Beg. Node Residual Pressure (psi)	Imbalance Flow At Beg. Node (gpm)
144	145	1	19.21					
145	144	1	-19.21	19.34	0.00	0.00	11.12	0.13261
150	15	1	-50.67	0.00	0.00	0.00	25.08	-0.21261
150	151	1	50.46					
151	150	1	-50.46	0.00	0.00	0.00	20.03	0.14362
151	152	1	25.94					
151	153	1	24.67					
152	151	1	-25.94	26.10	0.00	0.00	20.25	0.16497
153	151	1	-24.67	24.58	0.00	0.00	17.97	-0.08411
160	16	9	-99.53	0.00	0.00	0.00	20.67	-0.41913
160	161	1	53.72					
160	164	1	45.40					
161	160	1	-53.72	0.00	0.00	0.00	12.58	-0.04621
161	162	1	20.18					
161	167	1	33.49					
162	161	1	-20.18	20.27	0.00	0.00	12.21	0.09233
163	167	1	60.81	26.69	0.00	0.00	21.18	34.11980
164	160	1	-45.40	0.00	0.00	0.00	15.93	-0.10021
164	165	1	23.21					
164	166	1	22.08					
165	164	1	-23.21	23.34	0.00	0.00	16.19	0.12271
166	164	1	-22.08	22.19	0.00	0.00	14.64	0.10866
167	161	1	-33.49	0.00	0.00	0.00	8.52	0.00643
167	163	1	-60.81					
167	168	1	17.23					
168	167	1	-17.23	26.69	0.00	0.00	21.18	9.46561



Fire Sprinkler Output Data

Group Peak Pipe Output Data

Beg. End. Node	Nodal KFactor (K)	Elevation (feet)	Spk/Hose Discharge (gpm)	Residual Pressure (psi)	Nom. Dia. Inside Dia. C-Value	Q (gpm) Velocity (fps)	F. L./ft (psi/ft) Fittings Type-Grp	Pipe-Len. Fit-Len. Tot-Len. (ft)	PF-(psi) PE-(psi) PV-(psi)
1	0.00	0.00	0.00	126.31	8.00	740.73	0.00531	6.60	0.454
2	0.00	0.00	0.00	125.85	7.981	4.75	EBGC	79.00	0.000
	SCHED 40 WET STEEL				120		8	85.60	0.152
2	0.00	0.00	0.00	125.85	8.00	740.76	0.00531	3.00	0.111
3	0.00	-3.00	0.00	127.04	7.981	4.75	E	18.00	-1.299
	SCHED 40 WET STEEL				120		8	21.00	0.152
3	0.00	-3.00	0.00	127.04	8.00	740.72	0.00531	122.00	0.743
4	0.00	-3.00	0.00	126.30	7.981	4.75	E	18.00	0.000
	SCHED 40 WET STEEL				120		8	140.00	0.152
4	0.00	-3.00	0.00	126.30	8.00	740.74	0.00531	52.00	0.372
5	0.00	49.00	0.00	103.41	7.981	4.75	E	18.00	22.516
	SCHED 40 WET STEEL				120		8	70.00	0.152
5	0.00	49.00	0.00	103.41	8.00	740.75	0.00531	2.00	0.196
60	0.00	49.00	0.00	103.21	7.981	4.75	T	35.00	0.000
	SCHED 40 WET STEEL				120		8	37.00	0.152
7	0.00	49.00	0.00	103.09	5.00	640.66	0.03780	86.00	4.158
8	0.00	49.00	0.00	98.93	5.047	10.27	2E	24.00	0.000
	SCHED 40 WET STEEL				120		5	110.00	0.710
8	0.00	49.00	0.00	98.93	4.00	640.65	0.11365	127.00	14.433
9	0.00	49.00	0.00	84.50	4.026	16.15	----	0.00	0.000
	SCHED 40 WET STEEL				120		4	127.00	1.754
9	0.00	49.00	0.00	84.50	3.00	640.64	0.42687	25.00	20.063
10	0.00	49.00	0.00	64.44	3.068	27.80	ET	22.00	0.000
	SCHED 40 WET STEEL				120		3	47.00	5.202
10	0.00	49.00	0.00	64.44	1.50	193.86	1.08041	1.00	9.724
100	0.00	50.00	0.00	54.28	1.610	30.55	T	8.00	0.433
	SCHED 40 WET STEEL				120		9	9.00	6.281
10	0.00	49.00	0.00	64.44	2.50	446.75	0.63105	9.00	13.252
11	0.00	49.00	0.00	51.19	2.469	29.94	T	12.00	0.000
	SCHED 40 WET STEEL				120		3	21.00	6.031
11	0.00	49.00	0.00	51.19	1.50	103.60	0.33896	1.00	1.695
111	0.00	50.00	0.00	49.06	1.610	16.33	E	4.00	0.433
	SCHED 40 WET STEEL				120		9	5.00	1.794
11	0.00	49.00	0.00	51.19	2.50	343.14	0.38731	3.00	5.810
12	0.00	49.00	0.00	45.38	2.469	22.99	T	12.00	0.000
	SCHED 40 WET STEEL				120		3	15.00	3.558



Fire Sprinkler Output Data

Group Peak Pipe Output Data

Beg. End. Node	Nodal KFactor (K)	Elevation (feet)	Spk/Hose Discharge (gpm)	Residual Pressure (psi)	Nom. Dia. Inside Dia. C-Value	Q (gpm) Velocity (fps)	F. L./ft (psi/ft) Fittings Type-Grp	Pipe-Len. Fit-Len. Tot-Len. (ft)	PF-(psi) PE-(psi) PV-(psi)
12	0.00	49.00	0.00	45.38	1.00	74.49	1.48289	1.00	4.449
120	0.00	50.00	0.00	40.50	1.049	27.65	E	2.00	0.433
	SCHED 40 WET STEEL				120		1	3.00	5.146
12	0.00	49.00	0.00	45.38	2.00	268.65	0.58516	5.00	8.777
13	0.00	49.00	0.00	36.60	2.067	25.69	T	10.00	0.000
	SCHED 40 WET STEEL				120		2	15.00	4.440
13	0.00	49.00	0.00	36.60	1.00	58.50	0.94846	1.00	2.845
130	0.00	50.00	0.00	33.32	1.049	21.72	E	2.00	0.433
	SCHED 40 WET STEEL				120		1	3.00	3.174
13	0.00	49.00	0.00	36.60	2.00	210.21	0.37167	7.00	6.318
14	0.00	49.00	0.00	30.28	2.067	20.10	T	10.00	0.000
	SCHED 40 WET STEEL				120		2	17.00	2.718
14	0.00	49.00	0.00	30.28	1.00	60.28	1.00255	1.00	3.008
140	0.00	50.00	0.00	26.84	1.049	22.38	E	2.00	0.433
	SCHED 40 WET STEEL				120		1	3.00	3.370
14	0.00	49.00	0.00	30.28	2.00	150.05	0.19919	3.00	2.589
15	0.00	49.00	0.00	27.69	2.067	14.35	T	10.00	0.000
	SCHED 40 WET STEEL				120		2	13.00	1.385
15	0.00	49.00	0.00	27.69	1.00	50.67	0.72701	1.00	2.181
150	0.00	50.00	0.00	25.08	1.049	18.81	E	2.00	0.433
	SCHED 40 WET STEEL				120		1	3.00	2.381
15	0.00	49.00	0.00	27.69	1.50	99.19	0.31272	8.00	3.753
16	0.00	49.00	0.00	23.94	1.610	15.63	E	4.00	0.000
	SCHED 40 WET STEEL				120		9	12.00	1.644
16	0.00	49.00	0.00	23.94	1.50	99.53	0.31472	1.00	2.833
160	0.00	50.00	0.00	20.67	1.610	15.69	T	8.00	0.433
	SCHED 40 WET STEEL				120		9	9.00	1.656
60	0.00	49.00	0.00	103.21	8.00	640.72	0.00406	8.00	0.122
7	0.00	49.00	0.00	103.09	7.981	4.11	EG	22.00	0.000
	SCHED 40 WET STEEL				120		8	30.00	0.114
60	0.00	49.00	0.00	103.21	2.50	99.95	0.03953	73.00	2.886
61	0.00	49.00	0.00	100.33	2.469	6.70	----	0.00	0.000
	SCHED 40 WET STEEL				120		3	73.00	0.302
61	0.00	49.00	0.00	100.33	2.00	99.95	0.09394	54.00	5.542
62	0.00	49.00	0.00	94.79	2.067	9.56	E	5.00	0.000
	SCHED 40 WET STEEL				120		2	59.00	0.615



Fire Sprinkler Output Data

Group Peak Pipe Output Data

Beg. End. Node	Nodal KFactor (K)	Elevation (feet)	Spk/Hose Discharge (gpm)	Residual Pressure (psi)	Nom. Dia. Inside Dia. C-Value	Q (gpm) Velocity (fps)	F. L./ft (psi/ft) Fittings Type-Grp	Pipe-Len. Fit-Len. Tot-Len. (ft)	PF-(psi) PE-(psi) PV-(psi)
62	0.00	49.00	0.00	94.79	1.50	99.95	0.31716	67.00	22.518
63	0.00	49.00	0.00	72.27	1.610	15.75	E	4.00	0.000
SCHED 40 WET STEEL					120		9	71.00	1.670
63	0.00	49.00	0.00	72.27	1.50	99.95	0.31718	10.00	5.709
64	0.00	43.00	100.00	69.16	1.610	15.75	2E	8.00	-2.598
SCHED 40 WET STEEL					120		9	18.00	1.670
100	0.00	50.00	0.00	54.28	1.00	86.65	1.96144	6.00	21.576
101	0.00	50.00	0.00	32.71	1.049	32.17	T	5.00	0.000
SCHED 40 WET STEEL					120		1	11.00	6.963
100	0.00	50.00	0.00	54.28	1.50	107.20	0.36101	2.00	3.610
105	0.00	50.00	0.00	50.67	1.610	16.89	T	8.00	0.000
SCHED 40 WET STEEL					120		9	10.00	1.921
101	0.00	50.00	0.00	32.71	1.00	32.20	0.31422	4.00	1.885
103	0.00	50.00	0.00	30.82	1.049	11.95	E	2.00	0.000
SCHED 40 WET STEEL					120		1	6.00	0.962
101	0.00	50.00	0.00	32.71	1.00	54.44	0.83012	7.00	9.961
110	0.00	50.00	0.00	22.74	1.049	20.21	T	5.00	0.000
SCHED 40 WET STEEL					120		1	12.00	2.748
103	0.00	50.00	0.00	30.82	1.00	32.20	0.31422	1.00	0.314
104	5.80	49.00	32.26	30.94	1.049	11.95	----	0.00	-0.433
SCHED 40 WET STEEL					120		1	1.00	0.962
105	0.00	50.00	0.00	50.67	1.00	41.09	0.49331	1.00	0.493
106	5.80	49.00	41.26	50.61	1.049	15.25	----	0.00	-0.433
SCHED 40 WET STEEL					120		1	1.00	1.566
105	0.00	50.00	0.00	50.67	1.00	66.10	1.18880	9.00	16.643
107	0.00	50.00	0.00	34.03	1.049	24.54	T	5.00	0.000
SCHED 40 WET STEEL					120		1	14.00	4.052
107	0.00	50.00	0.00	34.03	1.00	33.77	0.34305	1.00	0.343
108	5.80	49.00	33.88	34.12	1.049	12.53	----	0.00	-0.433
SCHED 40 WET STEEL					120		1	1.00	1.057
107	0.00	50.00	0.00	34.03	1.00	32.25	0.31517	9.00	3.467
109	5.80	49.00	32.29	30.99	1.049	11.97	E	2.00	-0.433
SCHED 40 WET STEEL					120		1	11.00	0.965
110	0.00	50.00	0.00	22.74	1.00	26.69	0.22209	7.00	1.999
102	5.80	49.00	26.69	21.18	1.049	9.91	E	2.00	-0.433
SCHED 40 WET STEEL					120		1	9.00	0.661



Fire Sprinkler Output Data

Group Peak Pipe Output Data

Beg. End. Node	Nodal KFactor (K)	Elevation (feet)	Spk/Hose Discharge (gpm)	Residual Pressure (psi)	Nom. Dia. Inside Dia. C-Value	Q (gpm) Velocity (fps)	F. L./ft (psi/ft) Fittings Type-Grp	Pipe-Len. Fit-Len. Tot-Len. (ft)	PF-(psi) PE-(psi) PV-(psi)
110	0.00	50.00	0.00	22.74	1.00	27.74	0.23854	1.00	0.239
117	5.80	49.00	27.78	22.94	1.049	10.30	-----	0.00	-0.433
SCHED 40 WET STEEL					120		0	1.00	0.714
111	0.00	50.00	0.00	49.06	1.50	103.60	0.33894	2.00	3.389
112	0.00	50.00	0.00	45.67	1.610	16.33	T	8.00	0.000
SCHED 40 WET STEEL					120		9	10.00	1.794
112	0.00	50.00	0.00	45.67	1.00	39.03	0.44853	1.00	0.449
113	5.80	49.00	39.19	45.65	1.049	14.49	-----	0.00	-0.433
SCHED 40 WET STEEL					120		1	1.00	1.413
112	0.00	50.00	0.00	45.67	1.00	64.56	1.13801	7.00	13.656
114	0.00	50.00	0.00	32.01	1.049	23.97	T	5.00	0.000
SCHED 40 WET STEEL					120		1	12.00	3.865
114	0.00	50.00	0.00	32.01	1.00	32.95	0.32786	1.00	0.328
115	5.80	49.00	32.87	32.12	1.049	12.23	-----	0.00	-0.433
SCHED 40 WET STEEL					120		1	1.00	1.007
114	0.00	50.00	0.00	32.01	1.00	31.73	0.30576	6.00	2.446
116	5.80	49.00	31.77	30.00	1.049	11.78	E	2.00	-0.433
SCHED 40 WET STEEL					120		1	8.00	0.934
120	0.00	50.00	0.00	40.50	1.00	74.49	1.48292	6.00	16.312
121	0.00	50.00	0.00	24.18	1.049	27.65	T	5.00	0.000
SCHED 40 WET STEEL					120		1	11.00	5.146
121	0.00	50.00	0.00	24.18	1.00	27.82	0.23971	4.00	1.438
122	5.80	49.00	27.92	23.18	1.049	10.33	E	2.00	-0.433
SCHED 40 WET STEEL					120		1	6.00	0.718
121	0.00	50.00	0.00	24.18	1.00	46.58	0.62210	7.00	7.465
124	0.00	50.00	0.00	16.72	1.049	17.29	T	5.00	0.000
SCHED 40 WET STEEL					120		1	12.00	2.012
124	0.00	50.00	0.00	16.72	1.00	22.68	0.16428	9.00	1.807
123	5.80	49.00	22.72	15.34	1.049	8.42	E	2.00	-0.433
SCHED 40 WET STEEL					120		1	11.00	0.477
124	0.00	50.00	0.00	16.72	1.00	23.82	0.17992	1.00	0.180
125	5.80	49.00	23.89	16.97	1.049	8.84	-----	0.00	-0.433
SCHED 40 WET STEEL					120		1	1.00	0.526
130	0.00	50.00	0.00	33.32	1.00	58.49	0.94791	2.00	6.635
131	0.00	50.00	0.00	26.69	1.049	21.71	T	5.00	0.000
SCHED 40 WET STEEL					120		1	7.00	3.172



Fire Sprinkler Output Data

Group Peak Pipe Output Data

Beg. End. Node	Nodal KFactor (K)	Elevation (feet)	Spk/Hose Discharge (gpm)	Residual Pressure (psi)	Nom. Dia. Inside Dia. C-Value	Q (gpm) Velocity (fps)	F. L./ft (psi/ft) Fittings Type-Grp	Pipe-Len. Fit-Len. Tot-Len. (ft)	PF-(psi) PE-(psi) PV-(psi)
131	0.00	50.00	0.00	26.69	1.00	29.82	0.27260	1.00	0.273
132	5.80	49.00	30.05	26.85	1.049	11.07	-----	0.00	-0.433
	SCHED 40 WET STEEL				120		1	1.00	0.825
131	0.00	50.00	0.00	26.69	1.00	28.34	0.24814	10.00	2.978
133	5.80	49.00	28.50	24.14	1.049	10.52	E	2.00	-0.433
	SCHED 40 WET STEEL				120		1	12.00	0.745
140	0.00	50.00	0.00	26.84	1.00	60.24	1.00104	6.00	11.011
141	0.00	50.00	0.00	15.83	1.049	22.36	T	5.00	0.000
	SCHED 40 WET STEEL				120		1	11.00	3.365
141	0.00	50.00	0.00	15.83	1.00	22.54	0.16238	4.00	0.974
142	5.80	49.00	22.68	15.29	1.049	8.37	E	2.00	-0.433
	SCHED 40 WET STEEL				120		1	6.00	0.471
141	0.00	50.00	0.00	15.83	1.00	37.58	0.41818	7.00	5.018
144	0.00	50.00	0.00	10.81	1.049	13.95	T	5.00	0.000
	SCHED 40 WET STEEL				120		1	12.00	1.310
144	0.00	50.00	0.00	10.81	1.00	18.25	0.82791	10.00	9.935
143	5.80	49.00	26.69	21.18	1.049	6.77	E	2.00	-0.433
	SCHED 40 WET STEEL				120		1	12.00	0.309
144	0.00	50.00	0.00	10.81	1.00	19.21	0.12084	1.00	0.121
145	5.80	49.00	19.34	11.12	1.049	7.13	-----	0.00	-0.433
	SCHED 40 WET STEEL				120		1	1.00	0.342
150	0.00	50.00	0.00	25.08	1.00	50.46	0.72138	2.00	5.050
151	0.00	50.00	0.00	20.03	1.049	18.73	T	5.00	0.000
	SCHED 40 WET STEEL				120		1	7.00	2.361
151	0.00	50.00	0.00	20.03	1.00	25.94	0.21056	1.00	0.211
152	5.80	49.00	26.10	20.25	1.049	9.63	-----	0.00	-0.433
	SCHED 40 WET STEEL				120		1	1.00	0.624
151	0.00	50.00	0.00	20.03	1.00	24.67	0.19192	9.00	2.495
153	5.80	49.00	24.58	17.97	1.049	9.16	2E	4.00	-0.433
	SCHED 40 WET STEEL				120		1	13.00	0.564
160	0.00	50.00	0.00	20.67	1.00	53.72	0.80985	8.00	8.099
161	0.00	50.00	0.00	12.58	1.049	19.94	E	2.00	0.000
	SCHED 40 WET STEEL				120		1	10.00	2.676
160	0.00	50.00	0.00	20.67	1.00	45.40	0.59323	3.00	4.746
164	0.00	50.00	0.00	15.93	1.049	16.85	T	5.00	0.000
	SCHED 40 WET STEEL				120		1	8.00	1.911



Fire Sprinkler Output Data

Group Peak Pipe Output Data

Beg. End. Node	Nodal KFactor (K)	Elevation (feet)	Spk/Hose Discharge (gpm)	Residual Pressure (psi)	Nom. Dia. Inside Dia. C-Value	Q (gpm) Velocity (fps)	F. L./ft (psi/ft) Fittings Type-Grp	Pipe-Len. Fit-Len. Tot-Len. (ft)	PF-(psi) PE-(psi) PV-(psi)
161	0.00	50.00	0.00	12.58	1.00	20.18	0.13234	4.00	0.794
162	5.80	49.00	20.27	12.21	1.049	7.49	E	2.00	-0.433
	SCHED 40 WET STEEL				120		1	6.00	0.378
161	0.00	50.00	0.00	12.58	1.00	33.49	0.33793	7.00	4.055
167	0.00	50.00	0.00	8.52	1.049	12.43	T	5.00	0.000
	SCHED 40 WET STEEL				120		1	12.00	1.040
167	0.00	50.00	0.00	8.52	1.00	60.81	1.01883	10.00	12.226
163	5.80	49.00	26.69	21.18	1.049	22.58	E	2.00	0.433
	SCHED 40 WET STEEL				120		1	12.00	3.430
164	0.00	50.00	0.00	15.93	1.00	23.21	0.17152	1.00	0.172
165	5.80	49.00	23.34	16.19	1.049	8.62	----	0.00	-0.433
	SCHED 40 WET STEEL				120		1	1.00	0.500
164	0.00	50.00	0.00	15.93	1.00	22.08	0.15639	9.00	1.720
166	5.80	49.00	22.19	14.64	1.049	8.20	E	2.00	-0.433
	SCHED 40 WET STEEL				120		1	11.00	0.452
167	0.00	50.00	0.00	8.52	1.00	17.23	12.22590	1.00	12.226
168	5.80	49.00	26.69	21.18	1.049	6.39	----	0.00	-0.433
	SCHED 40 WET STEEL				120		1	1.00	0.275



Fire Sprinkler Output Data

Group Peak Sprinkler Output Data

Flowing Sprinkler Node No.	Area Group Code	Sprinkler KFactor (K)	Sprinkler Elevation (feet)	Residual Pressure (psi)	Flowing Area (ft²)	Flowing Density (gpm/ft²)	Sprinkler Discharge (gpm)
102		5.80	49.00	21.18	95.00	0.281	26.69
Sub Totals For Non-Group					95.00	0.281	26.69
104		5.80	49.00	30.94	45.00	0.717	32.26
Sub Totals For Non-Group					45.00	0.717	32.26
106		5.80	49.00	50.61	81.00	0.509	41.26
Sub Totals For Non-Group					81.00	0.509	41.26
108		5.80	49.00	34.12	81.00	0.418	33.88
Sub Totals For Non-Group					81.00	0.418	33.88
109		5.80	49.00	30.99	81.00	0.399	32.29
Sub Totals For Non-Group					81.00	0.399	32.29
113		5.80	49.00	45.65	81.00	0.484	39.19
Sub Totals For Non-Group					81.00	0.484	39.19
115		5.80	49.00	32.12	54.00	0.609	32.87
Sub Totals For Non-Group					54.00	0.609	32.87
116		5.80	49.00	30.00	44.00	0.722	31.77
Sub Totals For Non-Group					44.00	0.722	31.77
117		5.80	49.00	22.94	95.00	0.292	27.78
Sub Totals For Non-Group					95.00	0.292	27.78
122		5.80	49.00	23.18	45.00	0.621	27.92
Sub Totals For Non-Group					45.00	0.621	27.92
123		5.80	49.00	15.34	95.00	0.239	22.72
Sub Totals For Non-Group					95.00	0.239	22.72
125		5.80	49.00	16.97	95.00	0.252	23.89
Sub Totals For Non-Group					95.00	0.252	23.89
132		5.80	49.00	26.85	81.00	0.371	30.05
Sub Totals For Non-Group					81.00	0.371	30.05
133		5.80	49.00	24.14	86.00	0.331	28.50
Sub Totals For Non-Group					86.00	0.331	28.50
142		5.80	49.00	15.29	45.00	0.504	22.68
Sub Totals For Non-Group					45.00	0.504	22.68
143		5.80	49.00	21.18	95.00	0.281	26.69
Sub Totals For Non-Group					95.00	0.281	26.69
145		5.80	49.00	11.12	95.00	0.204	19.34
Sub Totals For Non-Group					95.00	0.204	19.34
152		5.80	49.00	20.25	81.00	0.322	26.10



Fire Sprinkler Output Data

Group Peak Sprinkler Output Data

Flowing Sprinkler Node No.	Area Group Code	Sprinkler KFactor (K)	Sprinkler Elevation (feet)	Residual Pressure (psi)	Flowing Area (ft²)	Flowing Density (gpm/ft²)	Sprinkler Discharge (gpm)
Sub Totals For Non-Group					81.00	0.322	26.10
153		5.80	49.00	17.97	86.00	0.286	24.58
Sub Totals For Non-Group					86.00	0.286	24.58
162		5.80	49.00	12.21	45.00	0.450	20.27
Sub Totals For Non-Group					45.00	0.450	20.27
163		5.80	49.00	21.18	95.00	0.281	26.69
Sub Totals For Non-Group					95.00	0.281	26.69
165		5.80	49.00	16.19	81.00	0.288	23.34
Sub Totals For Non-Group					81.00	0.288	23.34
166		5.80	49.00	14.64	148.00	0.150	22.19
Sub Totals For Non-Group					148.00	0.150	22.19
168		5.80	49.00	21.18	95.00	0.281	26.69
Sub Totals For Non-Group					95.00	0.281	26.69
Totals For All Groups					1925.00	0.348	669.66



Fire Sprinkler Output Summary

Hydraulically Most Demanding Sprinkler Node

HMD Sprinkler Node Number:	145
HMD Actual Residual Pressure:	11.12 psi
HMD Actual GPM:	19.34 gpm

Sprinkler Summary

Sprinkler System Type:	Wet
Specified Area Of Application:	2016.00 ft ²
Minimum Desired Density:	0.150 gpm/ft ²
Application Average Density:	0.332 gpm/ft ²
Application Average Area Per Sprinkler:	84.00 ft ²
Sprinkler Flow:	669.66 gpm
Average Sprinkler Flow:	27.90 gpm

Flow Velocity And Imbalance Summary

Maximum Flow Velocity (In Pipe 100 - 101)	32.17 ft/sec
Maximum Velocity Pressure (In Pipe 100 - 101)	6.96 psi
Allowable Maximum Nodal Pressure Imbalance:	0.2000 psi
Actual Maximum Nodal Pressure Imbalance:	0.1664 psi
Actual Average Nodal Pressure Imbalance:	0.0437 psi
Actual Maximum Nodal Flow Imbalance:	34.1198 gpm
Actual Average Nodal Flow Imbalance:	0.8535 gpm

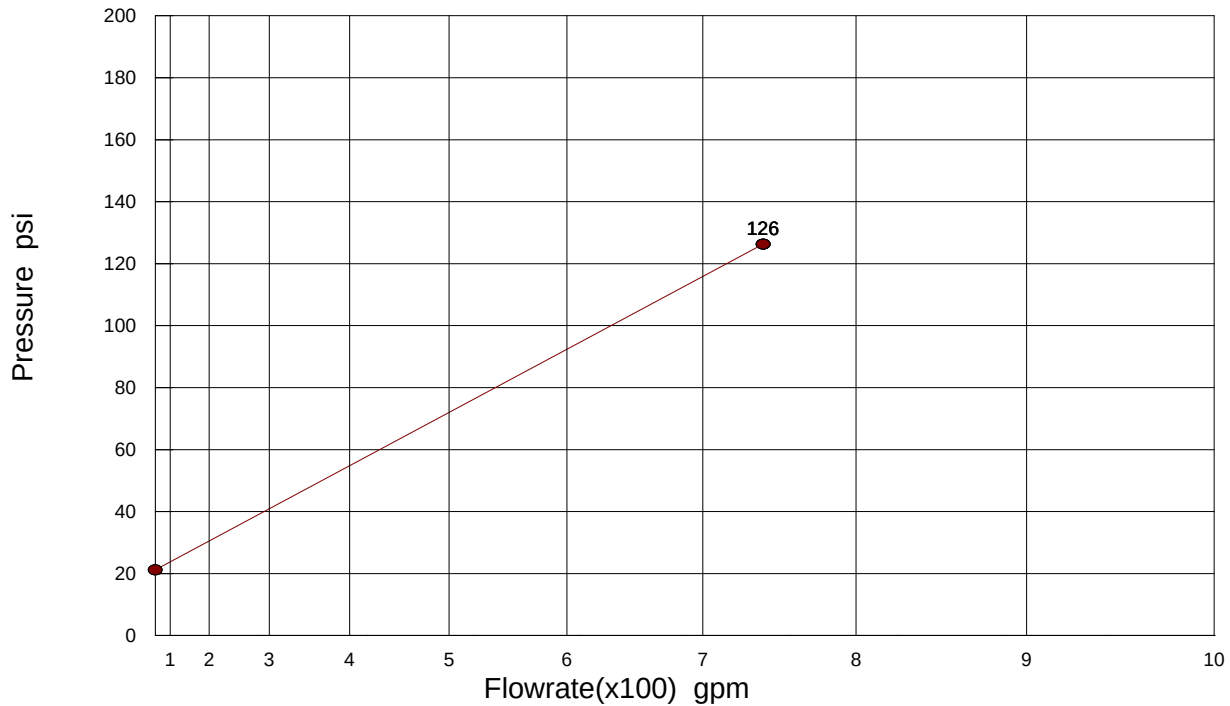
Overall Network Summary

Number Of Unique Pipe Sections:	66
Number Of Flowing Sprinklers:	24
Pipe System Water Volume:	737.43 gal
Sprinkler Flow:	669.66 gpm
Non-Sprinkler Flow:	100.00 gpm
Demand Flow Excluding Exterior Hose Flow:	769.66 gpm
Minimum Required Residual Pressure At System Inflow Node:	126.31 psi
Demand Flow At System Inflow Node:	740.73 gpm



Fire Sprinkler Output Data

Hydraulic Supply/Demand Graph



Demand Curve Data

Calculated Residual Pressure: 126.31 psi

Calculated Flow Rate: 740.73 gpm

Pressure Required For First Sprinkler Downstream From Inflow Node To Flow: 21.22 psi